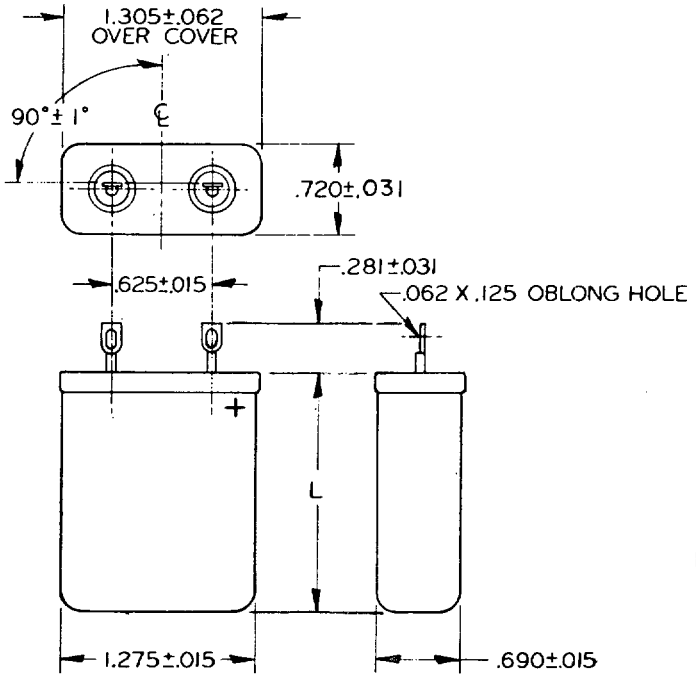
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG) - 85°C (VOLTAGE DERATED TO 125°C), STYLE CL55

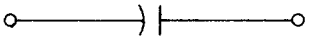
(E) INACTIVE FOR NEW EQUIPMENT DESIGN. NO SUPERSEDING DOCUMENT.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-3965.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



	Dimensions	
	L ±.031 (.79)	
	In	mm
A1	1.062	26.97
A2	1.375	34.93
A3	1.625	41.28
A4	2.000	50.80
A5	2.500	63.50



CIRCUIT DIAGRAM

INCHES	MM	INCHES	MM
.015	.38	.625	15.88
.031	.79	.690	17.53
.062	1.57	.720	18.29
.125	3.18	1.275	32.39
.281	7.14	1.305	33.15

NOTES:
1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Style CL55 capacitors.

(E) Denotes Change

FSC 5910

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TABLE I. Capacitor characteristics.

Type designation	Voltage			Capacitance	DC leakage (max)		Impedance (max)	Case size
	Rated (85°C)	Derated (125°C)	Surge (85°C)		25°C	85°C & 125°C		
	volts, dc	volts, dc	volts, dc	μF	μA	μA	ohms	
CL55BE961MPG	15	10	17.2	960	7	58	3.10	A1
CL55BE122MPG	15	10	17.2	1200	9	72	2.30	A2
CL55BE142MPG	15	10	17.2	1400	11	84	1.70	A3
CL55BE212MPG	15	10	17.2	2100	16	126	1.30	A4
CL55BE242MPG	15	10	17.2	2400	18	144	1.15	A5
CL55BH521MPG	30	20	34.5	520	8	63	5.30	A1
CL55BH661MPG	30	20	34.5	660	10	80	4.20	A2
CL55BH821MPG	30	20	34.5	820	13	99	2.90	A3
CL55BH122MPG	30	20	34.5	1200	18	144	2.30	A4
CL55BH132MPG	30	20	34.5	1300	20	156	2.10	A5
CL55BJ401MPG	50	30	57.5	400	10	80	7.20	A1
CL55BJ431MPG	50	30	57.5	430	10	78	6.40	A1
CL55BJ501MPG	50	30	57.5	500	13	100	5.60	A2
CL55BJ601MPG	50	30	57.5	600	15	120	4.10	A3
CL55BJ801MPG	50	30	57.5	800	20	160	3.10	A4
CL55BJ102MPG	50	30	57.5	1000	23	180	2.80	A5
CL55BL271MPG	75	50	86.2	270	9	81	8.50	A1
CL55BL331MPG	75	50	86.2	330	12	99	7.00	A2
CL55BL401MPG	75	50	86.2	400	15	119	5.00	A3
CL55BL601MPG	75	50	86.2	600	23	180	3.65	A4
CL55BL661MPG	75	50	86.2	660	25	198	3.50	A5
CL55BN171MPG	100	65	115.0	170	9	68	15.00	A1
CL55BN221MPG	100	65	115.0	220	11	88	11.60	A2
CL55BN261MPG	100	65	115.0	260	13	104	8.00	A3
CL55BN351MPG	100	65	115.0	350	18	140	6.50	A4
CL55BN441MPG	100	65	115.0	440	22	176	5.80	A5
CL55BQ700MPG	150	100	172.0	70	6	42	28.80	A1
CL55BQ900MPG	150	100	172.0	90	7	54	22.40	A2
CL55BQ101MPG	150	100	172.0	100	8	60	16.40	A3
CL55BQ141MPG	150	100	172.0	140	11	84	12.40	A4
CL55BQ181MPG	150	100	172.0	180	14	104	11.20	A5

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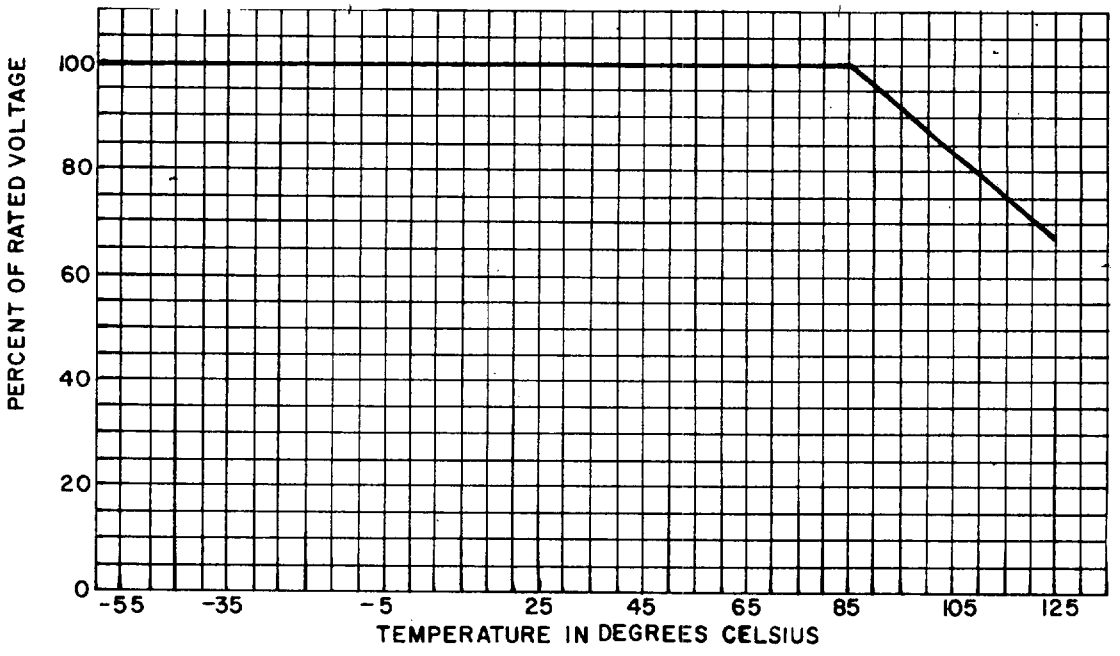


FIGURE 2. Voltage derating with temperature.

REQUIREMENTS:

Design and construction:
Dimensions and configuration:
Capacitors - See figure 1.
Brackets - See figure 3.
Case type - Rectangular, uninsulated.
Seal type - Hermetic (symbol G).
Terminals - See figure 1.
Rated temperature - -55° C to +85° C, voltage derated to +125° C (see figure 2).
Rated voltage - See table I.
Capacitance (cap) (nom) - See table I.
Capacitance tolerance - Symbol M (±20%).

DC leakage (DCL) (max) (at +25° C):
See table I.

Dissipation factor (DF) (max):
15% ($\leq$50 v) or 12% (>50 v).

Seal: In accordance with MIL-C-3965.

Dielectric withstanding voltage (DWV):
Sleeving to case (min) - Not applicable.
Terminal to case (min) - 300 volts.

Insulation resistance:
Sleeving to case (min) - Not applicable.
Terminal to case (min) - 100 megohms.

Temperature cycling:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±5% of initial measurement.
(E) DF (max) - 115% of initial requirement.

Solderability: Not applicable.

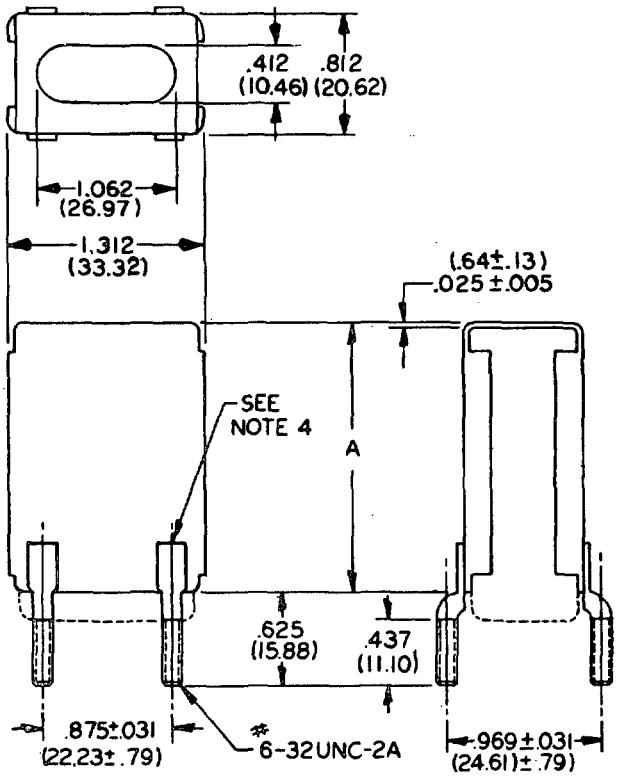
Terminal strength:
Pull test - In accordance with MIL-C-3965.
Bend test (if req.) - In accordance with MIL-C-3965.
Torque test - Not applicable.
Wire-lead bend test - Not applicable.
Twist test - Not applicable.

Surge voltage: See table I.

Moisture resistance:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±8% of initial measurement.
(E) DF (max) - 115% of initial requirement.

Low temperature (storage):
DCL (max) - See table I (25° C)
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

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BRACKETS FOR TYPE A CASE

Type designation	Case size	Dim	Dimensions				
			Inches			Millimeters	
			Min	Nom	Max	Min	Max
CL050SA13	A1	A	.846	.862	.878	21.49	22.30
	A2 1/	A	1.159	1.175	1.191	29.44	30.25
CL050SA33	A3	A	1.409	1.425	1.441	35.79	36.60
	A4 1/	A	1.784	1.800	1.816	45.31	46.13
	A5 1/	A	2.284	2.300	2.316	58.01	58.82

1/ CL050SD13, CL050SD23 and CL050SD33 brackets (of MIL-C-3965/9) may be used for case sizes A2, A4 and A5, respectively.

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Spade lugs shall be rigid (either double welded, double riveted, single riveted and soldered, or single riveted and welded to the bracket).
5. Unless otherwise specified, tolerance is ±.062 (1.57 mm).

FIGURE 3. CL050 brackets.

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Stability at low and high temperature:
Step 1 (+25° C):
DCL (max) - See table I.
Cap - Within tolerance of table I value.
DF (max) - See table I.
Step 2 (-55° C):
Impedance - See table I.
ΔCap (max) - -35% of step 1 value.
Step 3 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±5% of step 1 value.
DF (max) - Shall not exceed initial requirement.
Step 4 (+85° C):
DCL (max) - See table I.
ΔCap (max) - +25% of step 1 value.
DF (max) - Shall not exceed initial requirement.
Step 5 (+125° C):
DCL (max) - See table I.
ΔCap (max) - +25% of step 1 value.
DF (max) - Shall not exceed initial requirement.
Step 6 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Reverse voltage:

Life at +85° C:
2,000-hr (qualification):
DCL (max) - See table I (25° C).
ΔCap - Within ±25% of initial measurement.
DF (max) - 130% of initial requirement.
240-hr (group B):
DCL (max) - See table I (25° C).
ΔCap - Within ±10% of initial measurement.
DF (max) - 130% of initial requirement.
1,760-hr (group C):
DCL (max) - See table I (25° C).
ΔCap - Within ±25% of initial measurement.
DF (max) - 130% of initial requirement.

Barometric pressure:
Test voltage - Rated voltage.

Life at +125° C:
DCL (max) - See table I (25° C).
ΔCap - Within ±25% of initial measurement.
DF (max) - 130% of initial requirement.

Marking: Capacitors shall be marked as shown in the following example:
Example:
CL55BE961MPG-- Type designation.
12345 7015 - Source and date code.
960 μF 15 vdc - Capacitance and rated voltage (at 85° C).
Polarity - Plug (+) symbol near positive terminal (see figure 1).

Voltage group (85° C rated voltage):
I - - - - - - - - - - - 15 to 150 v, incl.

Substitution data. Capacitors covered by this specification are directly substitutable for capacitors of equivalent capacitance, capacitance tolerance, and voltage in the same case size.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU
Navy - SH
Air Force - 11, 17
DSA - ES

User activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Air Force - 80

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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